Infinite Series 8

Siddharth K

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Problem 1

Assume

$$\frac{p}{q} = [a_0; a_1, \ldots].$$

We want to show that it stops after finitely many terms.

Start with:

$$\frac{p}{q} = a_0 + \frac{r_1}{q}.$$

If $r_1 = 0$, the continued fraction has already terminated. If $r_1 > 0$,

$$\frac{p}{q} = a_0 + \frac{1}{q/r_1}.$$

Now divide q by r_1:

$$\frac{q}{r_1} = a_1 + \frac{r_2}{r_1},$$

$$\frac{p}{q} = a_0 + \frac{1}{a_1 + \frac{r_2}{r_1}}.$$

If we continue the process,

$$r_{k-1} = a_k r_k + r_{k+1}, \quad 0 \leq r_{k+1} < r_k,$$

since each remainder is a nonnegative integer and

$$q > r_1 > r_2 > \cdots \geq 0,$$

the sequence must hit 0, and the continued fraction stops.

This relates to the Euclidean algorithm because, when computing $\gcd(p, q)$,

$$\begin{aligned} p &= a_0 q + r_1, \\ q &= a_1 r_1 + r_2, \\ r_1 &= a_2 r_2 + r_3, \\ &\vdots \\ r_{n-2} &= a_n r_{n-1}. \end{aligned}$$

the quotients $a_0, \ldots, a_n$ are the continued-fraction coefficients.

Problem 2

Let $x = [a_0; a_1, \ldots, a_k, \overline{b_1, b_2, \ldots, b_n}]$ and $y = [\overline{b_1 : b_2, \ldots, b_n}]$. Let

$$x_i = [a_i; a_{i+1}, \ldots, a_k, \overline{b_1, b_2, \ldots, b_n}].$$

We claim that each x_i is of the form

$$x_i = \frac{ax + b}{cx + d}$$

for integers a, b, c, d. We can prove this by induction. We have that $x_0 = x$, so the base case holds. Now, assume

$$x_i = \frac{ax + b}{cx + d}.$$

We have that

$$x_{i+1} = \frac{1}{x_i - a_i} = \frac{1}{\frac{ax+b}{cx+d} - a_i} = \frac{cx + d}{ax + b - a_i(cx + d)} = \frac{cx + d}{(a - a_i c)x + b - a_i d}.$$

This means that x_k, and therefore y, can be written in the form $\frac{ax+b}{cx+d}$. By the same argument as before, y can be written as

$$y = \frac{py + q}{ry + s},$$

for integers p, q, r, s. So we have that

$$y(ry + s) = py + q,$$

$$ry^2 + (s - p)y - q = 0.$$

Substituting $y = \frac{ax+b}{cx+d}$,

$$r\left(\frac{ax + b}{cx + d}\right)^2 + (s - p)\left(\frac{ax + b}{cx + d}\right) + q = 0,$$

$$r(ax + b)^2 + (s - p)(ax + b)(cx + d) + q(cx + d)^2 = 0,$$

which is quadratic in x with integer coefficients. So x is a root of a quadratic polynomial with integer coefficients.

Problem 6

Let

$$y = \frac{x - 1}{2 + \frac{x-1}{2+\cdots}}.$$

Then

$$1 + y = 1 + \frac{x-1}{2+y}, \quad y^2 + 2y + 1 - x = 0,$$

$$y = \frac{-2 \pm \sqrt{4 - 4 + 4x}}{2} = -1 \pm \sqrt{x}.$$

Since y is positive,

$$y = -1 + \sqrt{x},$$

$$\sqrt{x} = y + 1,$$

which is the desired continued fraction.

Problem 7

Define

$$I_n = \int_0^1 \frac{x^n}{1+x} dx.$$

We have

$$I_n + I_{n+1} = \int_0^1 \left(\frac{x^n}{1+x} + \frac{x^{n+1}}{1+x} \right) = \int_0^1 x^n dx = \frac{1}{n+1},$$

$$\frac{I_{n+1} + I_n}{I_{n+2} + I_{n+1}} = \frac{n+2}{n+1}.$$

Define

$$r_n = \frac{I_{n+1}}{I_n}.$$

Then

$$\frac{1 + 1/r_n}{r_{n+1} + 1} = \frac{n+2}{n+1},$$

$$r_n = \frac{1}{\frac{n+2}{n+1}(r_{n+1}+1) - 1} = \frac{n+1}{n+1+(n+2)r_{n+1}}.$$

So

$$r_0 = \frac{1}{1+2r_1} = \frac{1}{1 + \dfrac{2^2}{1+3r_2}} = \cdots.$$

We can see that the desired continued fraction is $\frac{1}{1+r_0}$. Since $r_0 = \frac{I_1}{I_0}$, we have

$$I_0 = \int_0^1 \frac{1}{1+x} dx = \log(2),$$

$$I_1 = \int_0^1 \frac{x}{1+x} dx = 1 - \log(2),$$

$$\frac{1}{1+r_0} = \frac{I_0}{I_0 + I_1} = \log(2).$$